

Lesson 8-1: Understand Equations and Their Solutions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.5

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Evaluate $3x + 2$ when $x = 4$.

Answer: 14

Substitute and multiply first: $3 \times 4 + 2 = 14$.

Prerequisite skill: Substitute a value into an expression and evaluate

TASK 2 • FOUNDATION 2

Find the missing number: $7 + 5 = \underline{\quad} + 4$.

Answer: 8

Both sides must equal 12; $8 + 4 = 12$.

Prerequisite skill: Read an equals sign as a balance, not as “the answer is”

TASK 3 • FOUNDATION 3

Write an equation for a number n plus 6 is 15.

Answer: $n + 6 = 15$

Is connects two equal quantities, so use an equals sign.

Prerequisite skill: Translate a sentence into an equation

TASK 4 • FOUNDATION 4

Does $x = 5$ make $2x + 1 = 11$ true? Explain.

Answer: Yes; $2 \times 5 + 1 = 11$

Substitution gives 11 on the left, matching the right side.

Prerequisite skill: Compare two sides to decide whether they match

Lesson 8-1: Understand Equations and Their Solutions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.5

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Is $x = 4$ a solution of $3x = 12$? Write an equation for "three times a number is 15."

Answer: Substitute $x = 4$: $3(4) = 12$ is true, so YES, $x = 4$ is a solution. Equation: $3n = 15$ (or $3x = 15$).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Is $x = 5$ a solution of $4x = 20$? Write an equation for "five times a number is 35."

Answer: $4(5) = 20$ is true, so yes. Equation: $5n = 35$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Is $x = 3$ a solution of $2x + 1 = 8$? Write an equation for "a number increased by 9 is 21."

Answer: $2(3) + 1 = 7$, not 8, so no. Equation: $n + 9 = 21$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Which of 2, 4, or 6 is a solution of $3x - 2 = 10$?

Answer: $x = 4$: $3(4) - 2 = 10$. ✓ The others give 4 and 16.

Check the result against the situation and explain your reasoning.

Lesson 8-2: Write and Solve Equations Using Addition or Subtraction

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.7

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Find the missing number: $\underline{\quad} + 7 = 19$.

Answer: 12

Undo adding 7 by subtracting: $19 - 7 = 12$.

Prerequisite skill: Recall addition and subtraction facts and their inverses

TASK 2 • FOUNDATION 2

Compute $5.2 - 1.85$.

Answer: 3.35

Write 5.20 and line up decimal places before subtracting.

Prerequisite skill: Add and subtract multi-digit numbers, decimals, and fractions

TASK 3 • FOUNDATION 3

Which operation undoes subtracting 8?

Answer: Adding 8

Adding 8 and subtracting 8 are inverse operations.

Prerequisite skill: Name the inverse of adding or subtracting a number

TASK 4 • FOUNDATION 4

Check $x = 9$ in $x + 6 = 15$. State the left-side value.

Answer: 15

Substitute 9: $9 + 6 = 15$, matching the right side.

Prerequisite skill: Substitute the answer back in to check

Lesson 8-2: Write and Solve Equations Using Addition or Subtraction

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.7

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$m - 11 = 25$, so $m = ?$ What undoes "+ 8"?

Answer: $m - 11 = 25 \rightarrow m = 25 + 11 = 36$. What undoes '+ 8': Subtracting 8 (or '- 8').

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$m - 14 = 31$, so $m = ?$ What undoes "+ 12"?

Answer: $m = 31 + 14 = 45$. Subtracting 12 undoes adding 12.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$y + 18 = 40$, so $y = ?$ What undoes "- 7"?

Answer: $y = 40 - 18 = 22$. Adding 7 undoes subtracting 7.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A jar lost 17 marbles and now holds 48. Write and solve an equation for the original count.

Answer: $m - 17 = 48$, so $m = 65$. Check: $65 - 17 = 48$. ✓

Check the result against the situation and explain your reasoning.

Lesson 8-3: Write and Solve Equations Using Multiplication or Division

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.7

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute 9×7 .

Answer: 63

Use $10 \times 7 - 7 = 63$.

Prerequisite skill: Recall multiplication and division facts

TASK 2 • FOUNDATION 2

Compute $12.6 \div 3$.

Answer: 4.2

126 tenths divided by 3 is 42 tenths.

Prerequisite skill: Divide multi-digit numbers and decimals

TASK 3 • FOUNDATION 3

Which operation undoes multiplying by 6?

Answer: Dividing by 6

Division by the same nonzero factor reverses multiplication.

Prerequisite skill: Name the inverse of multiplying or dividing by a number

TASK 4 • FOUNDATION 4

Compute $12 \div 3/4$.

Answer: 16

Multiply by the reciprocal: $12 \times 4/3 = 16$.

Prerequisite skill: Divide by a fraction to undo multiplying by one

Lesson 8-3: Write and Solve Equations Using Multiplication or Division

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.7

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$6x = 42$, so $x = ?$ $x \div 5 = 9$, so $x = ?$ What undoes " $\times 4$ "?

Answer: $6x = 42 \rightarrow x = 42 \div 6 = 7$. $x \div 5 = 9 \rightarrow x = 9 \times 5 = 45$. What undoes ' $\times 4$ ': Dividing by 4 (or ' $\div 4$ ').

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$8x = 56$, so $x = ?$ $x \div 4 = 7$, so $x = ?$ What undoes " $\times 9$ "?

Answer: $x = 56 \div 8 = 7$. $x = 7 \times 4 = 28$. Dividing by 9 undoes multiplying by 9.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$5x = 45$, so $x = ?$ $x \div 3 = 12$, so $x = ?$ What undoes " $\div 6$ "?

Answer: $x = 45 \div 5 = 9$. $x = 12 \times 3 = 36$. Multiplying by 6 undoes dividing by 6.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Tickets cost the same amount. Nine tickets cost \$63. Write and solve an equation for one ticket.

Answer: $9t = 63$, so $t = \$7$. Check: $9(7) = 63$.
✓

Check the result against the situation and explain your reasoning.

Lesson 8-4: Write and Represent Inequalities

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.8

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Which symbol means greater than or equal to?

Answer: $\geq$

The line under $>$ includes equality at the boundary.

Prerequisite skill: Use $<$, $>$, $\leq$, and $\geq$ correctly

TASK 2 • FOUNDATION 2

Write an inequality for a number n that is at least 12.

Answer: $n \geq 12$

At least includes 12 and every larger value.

Prerequisite skill: Translate "at least," "at most," "no more than," "more than"

TASK 3 • FOUNDATION 3

On a number line marked in ones, what number is two steps left of 1?

Answer: -1

Move from 1 to 0, then to -1.

Prerequisite skill: Place a number on a number line

TASK 4 • FOUNDATION 4

Does 7 satisfy $n < 5$? Explain.

Answer: No; 7 is greater than 5

Substituting 7 gives $7 < 5$, which is false.

Prerequisite skill: Decide whether a given value makes a comparison true

Lesson 8-4: Write and Represent Inequalities

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.8

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Write an inequality for "you must be at least 48 inches tall." Is 50 a solution?

Answer: Inequality: $h \geq 48$. Is 50 a solution? Yes, because $50 \geq 48$ is true.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Write an inequality for "you may bring at most 3 guests." Is 3 a solution?

Answer: $g \leq 3$. Yes — $3 \leq 3$ is true.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Write an inequality for "the ride requires more than 40 inches." Is 40 a solution?

Answer: $h > 40$. No — $40 > 40$ is false.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A bus holds no more than 44 riders and already has 31. Write an inequality for how many more can board.

Answer: $31 + r \leq 44$, so $r \leq 13$.

Check the result against the situation and explain your reasoning.

Lesson 8-5: Understand Inequalities and Their Solutions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.5

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Describe the graph of $x > 3$ on a number line.

Answer: Open circle at 3; shade right

Exclude 3 because the symbol is strict; values greater than 3 lie right.

Prerequisite skill: Mark a boundary point on a number line and shade one direction

TASK 2 • FOUNDATION 2

For $x \leq 4$, is the circle at 4 open or closed?

Answer: Closed

The equality part includes the boundary value 4.

Prerequisite skill: Choose an open or closed circle from the symbol

TASK 3 • FOUNDATION 3

Which of 2, 5, 8 satisfy $x \geq 5$?

Answer: 5 and 8

Both are at least 5; 2 is below the boundary.

Prerequisite skill: Substitute and test several candidate values

TASK 4 • FOUNDATION 4

Which is greater: -2 or -6?

Answer: -2

-2 is farther right on the number line.

Prerequisite skill: Compare a value against the boundary number

Lesson 8-5: Understand Inequalities and Their Solutions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.5

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Graph $x \geq 3$. Open circle or closed? Is 3 itself a solution?

Answer: Graph $x \geq 3$: Draw a solid closed circle at 3 and shade an arrow pointing to the right. Circle type: Closed circle. Is 3 a solution? Yes, because $3 \geq 3$ is true.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Graph $x > 5$. Open circle or closed? Is 5 itself a solution?

Answer: Open circle at 5, shade to the right. No — $5 > 5$ is false.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Graph $x \leq -2$. Open circle or closed? Is -2 itself a solution?

Answer: Closed circle at -2 , shade to the left. Yes — $-2 \leq -2$ is true.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Which of -1 , 0 , 4 , and 4.5 satisfy $x \geq 4$?

Answer: 4 and 4.5. The values -1 and 0 are to the left of the boundary.

Check the result against the situation and explain your reasoning.

Lesson 8-6: Solve and Graph Inequalities

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.8

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Solve $x + 7 = 19$.

Answer: 12

Subtract 7 from both sides: $x = 19 - 7 = 12$.

Prerequisite skill: Solve a one-step equation with an inverse operation

TASK 2 • FOUNDATION 2

Read $x \geq 4$ in words.

Answer: x is greater than or equal to 4.

$\geq$ means greater than or equal to.

Prerequisite skill: Read the symbols $<$, $>$, $\leq$ and $\geq$ in words

TASK 3 • FOUNDATION 3

Is 6 a solution of $x < 6$? Explain.

Answer: No. $6 < 6$ is false, because 6 equals 6.

The boundary of a strict inequality is not a solution.

Prerequisite skill: Test whether a value makes an inequality true by substitution

TASK 4 • FOUNDATION 4

A number line has tick marks every 5 units, starting at 0. What number sits 4 tick marks to the right of 0?

Answer: 20

Each tick adds 5: $4 \times 5 = 20$.

Prerequisite skill: Locate whole numbers on a scaled number line

Lesson 8-6: Solve and Graph Inequalities

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.8

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Solve $x + 5 = 13$. Is 4 a solution of $x \leq 4$? Read $x > 9$ in words.

Answer: $x = 13 - 5 = 8$. Yes: $4 \leq 4$ is true. x is greater than 9.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Solve $x - 6 = 10$. Is 7 a solution of $x > 7$? Read $x \leq 3$ in words.

Answer: $x = 16$. No: $7 > 7$ is false. x is less than or equal to 3.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Solve $x + 11 = 20$. Is 2 a solution of $x \geq 1$? Read $x < 15$ in words.

Answer: $x = 9$. Yes: $2 \geq 1$ is true. x is less than 15.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A ride needs riders at least 48 inches tall. Mia is 41 inches tall. Write and solve an inequality for how many more inches g she must grow.

Answer: $41 + g \geq 48$, so $g \geq 7$. She must grow at least 7 more inches.

Check the result against the situation and explain your reasoning.

Lesson 8-7: Equations and Inequalities Problem Solving

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.7

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Write "at least 30" as an inequality for m minutes.

Answer: $m \geq 30$

At least 30 means 30 or more.

Prerequisite skill: Translate at most, at least, more than and fewer than into inequality symbols

TASK 2 • FOUNDATION 2

Solve $p + 17 = 52$.

Answer: 35

Subtract 17 from both sides: $p = 52 - 17 = 35$.

Prerequisite skill: Solve a one-step equation with addition or subtraction

TASK 3 • FOUNDATION 3

Solve $6w = 54$.

Answer: 9

Divide both sides by 6: $w = 54 \div 6 = 9$.

Prerequisite skill: Solve a one-step equation with multiplication or division

TASK 4 • FOUNDATION 4

Is 9 a solution of $4x = 32$? Show the check.

Answer: No. $4 \times 9 = 36$, not 32. The solution is $x = 8$.

Substitute and compare both sides.

Prerequisite skill: Check a solution by substituting it into the original equation

Lesson 8-7: Equations and Inequalities Problem Solving

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.7

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Solve $m + 15 = 40$. Solve $7k = 63$. Write "fewer than 10" for b .

Answer: $m = 40 - 15 = 25$. $k = 63 \div 7 = 9$. $b < 10$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Solve $y - 8 = 21$. Solve $9z = 72$. Write "no more than 12" for c .

Answer: $y = 29$. $z = 8$. $c \leq 12$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Solve $a + 26 = 50$. Solve $4d = 52$. Write "at least 18" for h .

Answer: $a = 24$. $d = 13$. $h \geq 18$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A phone plan allows at most 500 texts a month. Ava sends 35 a day for 12 days. Can she send 80 more this month? Explain with an inequality.

Answer: $35 \times 12 = 420$. $420 + 80 = 500$, and $500 \leq 500$ is true, so yes, exactly at the limit.

Check the result against the situation and explain your reasoning.